ScanProTech: Compact mmWave Imager

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GitHub Demo

PROBLEM STATEMENT

Passive millimeter-wave (mmWave) imaging is a powerful technique for non-invasive detection in fields such as airport security, healthcare diagnostics, and industrial monitoring. However, most commercial systems are bulky, expensive, and do not support real-time data processing and interpretation.

ScanProTech aims to address this by developing an intuitive, compact, affordable mmWave imager capable of scanning objects and environments and classifying results automatically with the aid of Artificial Intelligence. The system leverages ambient thermal emissions in the W-band spectrum (90–100 GHz) to form passive images and identify potential threats or anomalies without the use of harmful radiation.

GOALS and REQUIREMENTS

This project aims to build a portable mmWave imager for detecting concealed threats and anomalies using passive radiation. This proof-of-concept prototype is intended for applications in security screening, industrial inspection, and biomedical imaging.

Key Objectives:

•Achieve high-resolution 20×20 pixel mmWave imaging using a synchronized dual-axis motion system.
•Enable real-time image generation through Python-based data acquisition and GUI control.
•Ensure portability and a compact form factor with minimal onboard computing requirements.
•Integrate AI-driven analysis to assess scan sharpness, classify objects, and identify potential threats.
•Provide intuitive, user-friendly visual feedback and diagnostic insights.
•Deliver a cost-effective alternative to existing industrial imaging systems

APPROACH

Our system utilizes a W-band Farran radiometer as the primary sensor, paired with an aspherical Rexolite lens to focus thermal radiation. The radiometer is mounted on a two-axis linear actuator platform driven by NEMA-17 stepper motors and Adafruit DC & stepper-motor bonnets.
A Raspberry Pi 5 single-board computer is used for scan control and data acquisition, while heatmaps are generated in real-time using Python libraries including Seaborn. Each scan is followed by the following data processing mechanisms:

- **Signal conditioning via the baseband circuit**, which includes a band-pass filter, low-noise amplifier, and integrator stages that clean and amplify the radiometer signal prior to digitization
- AI classification using a pre-trained CNN (ResNet18) and Heuristic evaluation for the scan produced

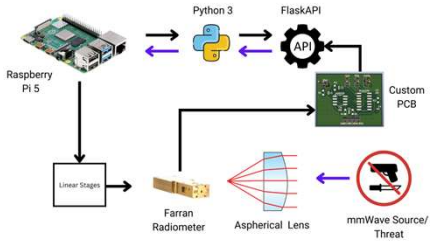
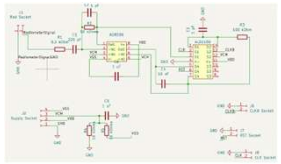


Fig. 1: Flowchart of the complete system pipeline, from mmWave sensing and motor control to real-time heatmap rendering and AI analysis.

DESIGN

GUI Intentions:
A Python-based GUI built with the Kivy framework provides intuitive control over the scanning system. It enables users to set parameters, initiate scans, monitor motor positions and velocities, and visualize results through real-time ADC acquisition and automatic heatmap display along with AI heuristic feedback. Initially, a web-based GUI was considered to broaden accessibility; however, a Python-native interface provided smoother integration with the system's hardware and backend processes, as shown in Figure. 7.

Baseband Circuit:
A three-part chopping circuit design that takes in the signal from the radiometer and passes it through a band-pass filter/low-noise amplifier. Designed to peak around 200 kHz, the circuit controls around 17dB of gain and filters the surrounding noise. The output is taken to the mixer to pass the signal with clock signals to improve the signal, as well as to pass the DC component. This DC signal gets taken in by the integrating circuit part, which further filters the circuit and gives the desired output, which is used for information and heatmap generation. Implemented are also bypass capacitors to reduce noise, and for the PCB, the copper pour with vias. The copper is connected to the ground plane to help maintain signal integrity and prevent outside interference with signals. The PCB copper layers are routing, GND, split supply, and routing.



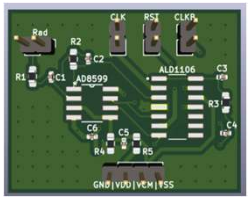


Fig. 2: Custom PCB Design Schematic and Layout

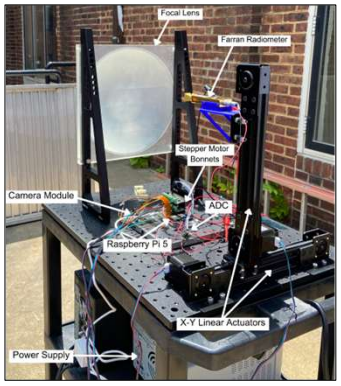


Fig. 3: Experimental Setup for the mmWave Scanner



Fig. 4: Displaying custom casing prototype along with aspherical lens

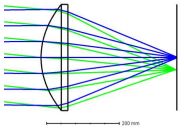


Fig. 5: Aspherical lens provide a single centric focus ray detection

Portability and Durability:
To enhance durability and usability in various environments, the system is housed within a custom-built protective casing. The structure consists of 12 aluminum T-slot extrusions forming a box frame, with transparent acrylic panels enclosing all sides except the front face. Then front is open for Radiometer lens access allowing a single-point ray detection via aspherical lens

Raspberry Pi Processing:
The Raspberry Pi 5 serves as the central unit for real-time scanning, data acquisition, and signal processing. After each scan, a heatmap is generated based on mmWave signal intensities, and analyzed using a **locally hosted FastAPI backend** that performs:

- **Pre-trained CNN (ResNet18) model:** Classifies object types using mmWave heatmap distributions, learning spatial patterns from annotated training data.
- **Heuristic AI Analyzer:** A lightweight, rule-based model that evaluates the heatmap based on brightness and Laplacian sharpness metrics to infer likely object type, confidence score (0.1 and 0.75), and extracts visual sharpness indicators.
- **Backend via FastAPI:** Operates as a REST API at *127.0.0.1:8000*, enabling asynchronous image POST requests from the frontend and ensuring efficient communication and low-latency response for GUI feedback integration
- **Modular Architecture and Extensible Design:** Easily upgradable—future models (e.g., improved CNNs) can be swapped for scalability and performance without modifying the GUI or scan pipeline.

COMPONENT and SYSTEM TESTING

Circuit Testing:
A sine wave was passed through the 1st circuit to test gain, noise, and filtering. A frequency sweep was also performed to get real values for amplification. For the mixer, 2 clock signals were measured for accuracy, and a sine was passed at a lesser frequency to verify the shape of the output clock and sine. Finally, the integrator was given a DC signal and passed a sawtooth, proving its success. After working around the arising issues, the combined circuit outputted the desired shape. It was given a square wave and output a sawtooth signal.

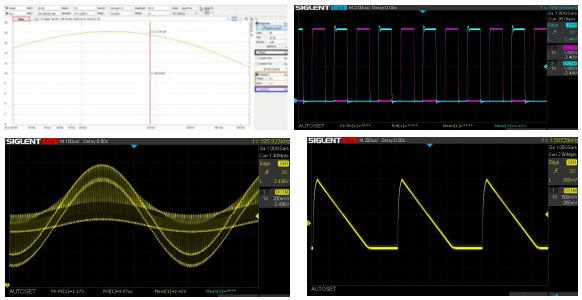


Fig. 6: Circuit Testing Graph Output

GUI Scan and Testing:
The GUI is displayed below showcasing parameters for scan control, including setting sampling rates, initiating scans, live ADC and velocity updates, and seamless heatmap rendering. Additionally, AI feedback—delivered via the backend API—was validated to return consistent object classification, threat scores, and captions in real time, enhancing overall system intelligence.

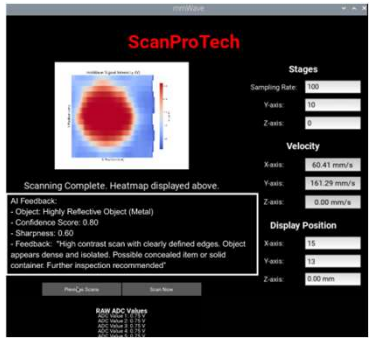


Fig. 7: ScanProTech Graphical User Interface

PROJECT STATUS

Current Status:
The mmWave imager reliably performs real-time scanning, heatmap generation, and AI-based analysis. The GUI enables users to adjust scan parameters, visualize results, and receive object feedback via a FastAPI backend.

Future Work:

- Integrating a deep learning classifier for improved object detection accuracy
- Improving upon PCB metrics for sharper image processing
- Deploy backend on edge AI hardware (e.g., Jetson Orin) for scalability
- Improving thermal and mechanical robustness for field deployment